ELEC-ACC-01 Access Node Hardware/Environmental Fault Resolution SOP (Arista CLI Dialect)

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3749

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults on the ELEC-ACC-01 access node, specifically related to ASIC memory parity errors, line card crashes, and power supply redundancy loss. The ELEC-ACC-01 node, located at the Electronic City Hub, is an Arista access node operating in the Access role. The failure domain for this node is classified as a Hardware/Environmental Fault (Severity: Warning), with an Alarm Code of ALARM_3749. This SOP is tailored to the specific requirements of the Arista CLI dialect and aims to ensure efficient and effective resolution of the failure domain.

Pre-requisite Checks

- Check: Verify redundant link status
- Check: Perform environmental checks (temperature, humidity, power supply)
- Check: Dispatch remote hands to the Electronic City Hub for physical access

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resources, Expected Output: System resources utilization and memory availability, Description: This command provides an overview of the system resources utilized by the node, including memory, CPU, and interface utilization, which can help identify potential memory-related issues.

Command: show interfaces, Expected Output: Interface status and error counters, Description: This command displays the status of all interfaces on the node, including any error counters that may indicate line card crashes or other hardware-related issues.

Command: show power redundancy, Expected Output: Power supply redundancy status, Description: This command verifies the status of the power supply redundancy, which can help identify potential power-related issues that may be contributing to the failure domain.
```

Resolution Steps

Step 1: Step: Replace faulty ASIC memory, Command: replace system memory, Description: This step involves replacing the faulty ASIC memory to resolve the memory parity error. The Arista CLI command 'replace system memory' is used to initiate the replacement process.

Step 2: Step: Replace faulty line card, Command: replace module, Description: This step involves replacing the faulty line card to resolve the line card crash issue. The Arista CLI command 'replace module' is used to initiate the replacement process.

Step 3: Step: Verify power supply redundancy, Command: show power redundancy, Description: This step involves verifying the status of the power supply redundancy to ensure it is functioning correctly and not contributing to the failure domain.

Post-Check Verification

```
Command: show system resources, Expected Output: System resources utilization and memory availability, Description: This command is used to verify that the memory parity error has been resolved and the system resources are functioning correctly.

Command: show interfaces, Expected Output: Interface status and error counters, Description: This command is used to verify that the line card crash has been resolved and the interface status is correct.

Command: show power redundancy, Expected Output: Power supply redundancy status, Description: This command is used to verify that the power supply redundancy is functioning correctly and not contributing to the failure domain.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy running-config startup-config, Description: This step involves restoring the original configuration of the node to its previous state. The Arista CLI command 'copy running-config startup-config' is used to initiate the restoration process.

- Step: Revert changes, Command: rollback, Description: This step involves reverting any changes made during the resolution process to ensure the node returns to its original state.